

WASP-36b

R-1588

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_220104 · created 2026-08-02T00:41:31+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459583.9057 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1669 +/- 0.0053
Transit depth (Rp/Rs)^2	2.79 +/- 0.18 [%]
Orbital Inclination (inc)	84.62 +/- 1.19
Airmass coefficient 1 (a1)	1.347 +/- 0.023
Airmass coefficient 2 (a2)	-0.206 +/- 0.035
Scatter in the residuals of the lightcurve fit is	1.72 %
Best Comparison Star	None
Optimal Aperture	3.54
Optimal Annulus	10.06
Transit Duration (day)	0.0863 +/- 0.0056

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1669 $\pm$ 0.0053 vs 0.1368 (deviation 3.64 σ)
Duration fit vs published	0.0863 d vs 0.0758 d (deviation 1.2 σ)
Mid-time O-C	9.0 min
Depth SNR	20.2
Residual scatter	1.72 %

Light curve

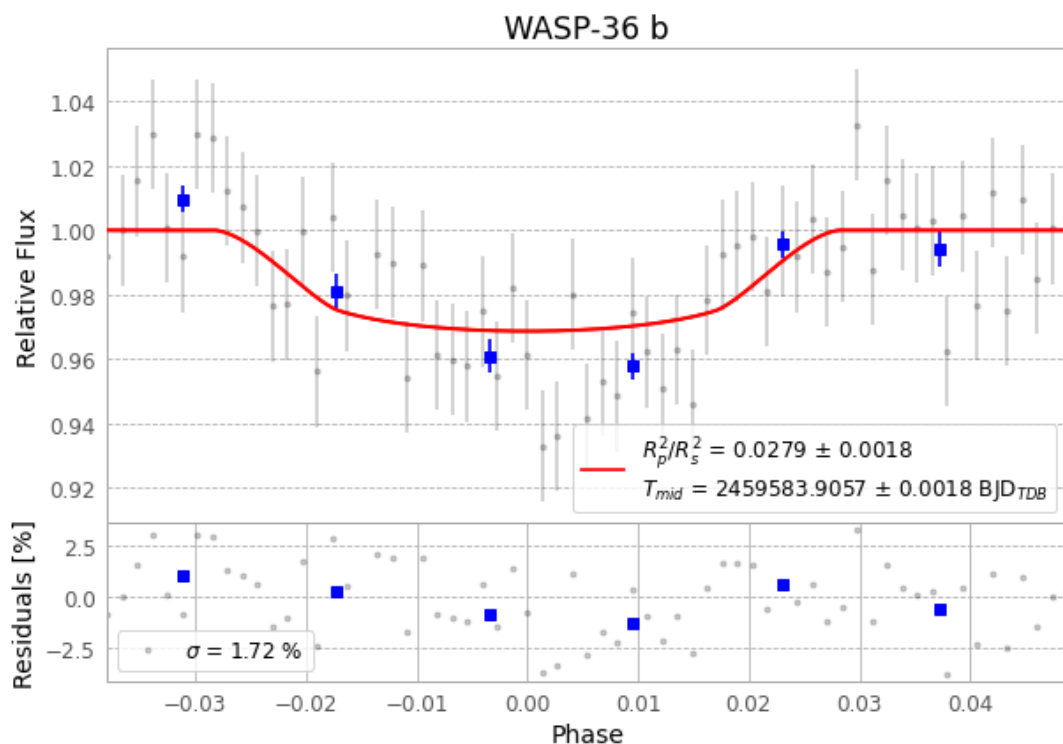


Figure 1 — FinalLightCurve_WASP-36 b_2022-01-04.png (SHA-256 19521d74619c2094...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [407, 172], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9eb23b4877465349...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

65 FITS frames (43 MB), WASP-36220104081213.FITS → WASP-36220104112413.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-220104.log (6c702755aad0...), FinalLightCurve_WASP-36 b_2022-01-04.png (19521d74619c...), FinalLightCurve_WASP-36 b_2022-01-04.csv (d08e717d810d...)

Compute resources

Wall time 155.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 00:41Z.